

ResPulse

Performance Regression Detection and Explanation

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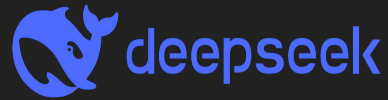
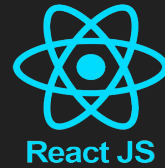
Problem and Motivation

- Performance regressions happen when a system becomes slower after a change, even if the code still builds and functions correctly.
- In distributed systems like ResilientDB, these regressions can be difficult to notice because latency and throughput changes may appear gradually or only under certain workloads.
- Existing build/test workflows can confirm correctness, but they do not always explain whether performance has degraded.
- This motivated **ResPulse**: a monitoring tool designed to detect silent performance regressions and provide readable explanations.

Project Overview

- Runs automated benchmark tests against ResilientDB transaction endpoints.
- Collects key performance metrics such as latency, throughput, success rate, and error behavior.
- Compares current benchmark results against historical baseline performance.
- Detects possible regressions when performance drops beyond expected thresholds.
- Generates AI-assisted explanations and recommendations for developers.
- Displays results through a dashboard and alert system.

Implementation Stack



System Overview

Performance Testing Engine

- Bash-based, uses curl's native timing capabilities to capture timing at every layer
- Captures: TCP connect time, server processing time, response transfer time, end-to-end latency



```
1
2  for i in $(seq 1 $RUNS); do
3      RESULT=$(curl -s -o /dev/null \
4          -w "%{http_code} %{time_connect} %{t
5             ime_pretransfer} %{time_starttransfer} %{
6                 time_total}" \
7             -X POST \
8             -H "Content-Type: application/json"
9             \
10             -d "{\"id\":\"perfkey$i\",\"value
11                 \":\"perfval$i\"}" \
```

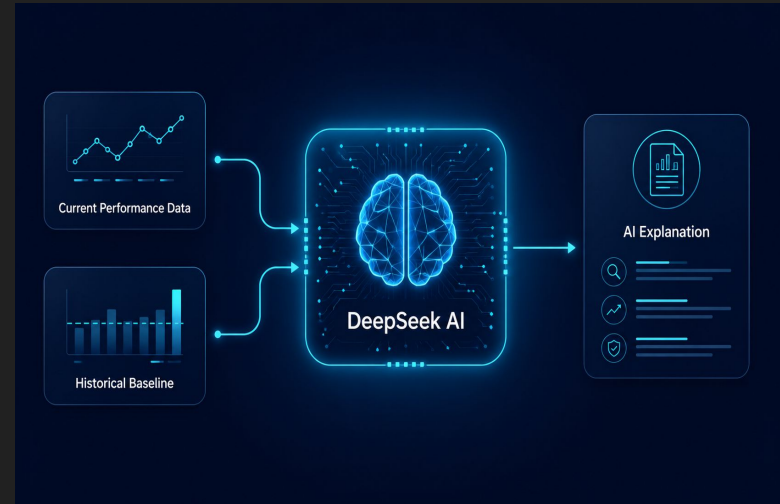
Data Analysis Engine

- Computes 6 core metrics per run: Total Latency, Consensus Time, TCP Connect Time, Transfer Time, Throughput (req/s), Success Rate
- Full statistical summaries: mean, median, min, max, standard deviation, P95 and P99 percentiles for tail latency detection
- Compares each run against a historical baseline (24h, 7d, 30d, 6m, all-time)

```
"runs": 500,  
"success_rate": 100,  
"throughput_rps": 2705.22,  
"avg_latency_ms": 0.37,  
"total_latency": {  
  "mean": 0.37,  
  "min": 0.21,  
  "max": 6.91,  
  "stddev": 0.36,  
  "p50": 0.31,  
  "p95": 0.6,  
  "p99": 1.23  
},  
"consensus_time_ms": {  
  "mean": 0.26,  
  "min": 0.14,  
  "max": 5.69,  
  "stddev": 0.29,  
  "p50": 0.21,  
  "p95": 0.39,  
  "p99": 0.75  
},  
"tcp_connect_ms": {  
  "mean": 0.06,  
  "min": 0.03,  
  "max": 0.87,  
  "stddev": 0.05,  
  "p50": 0.06,  
  "p95": 0.09,  
  "p99": 0.14  
},  
"transfer_time_ms": {  
  "mean": 0.03,  
  "min": 0.01,  
  "max": 0.64,  
  "stddev": 0.04,  
  "p50": 0.02,  
  "p95": 0.08,  
  "p99": 0.14  
},  
"version": ""
```

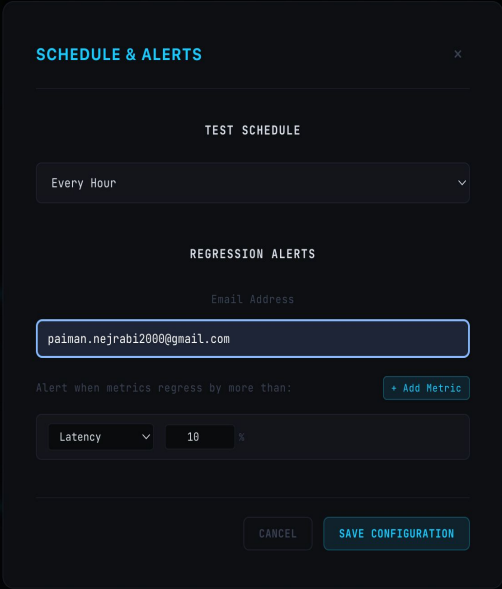
AI-powered analysis

- Called on-demand when the user clicks "Generate Analysis"
- The AI receives the current run metrics, the baseline, and PBFT-specific context
- Returns a human-readable report: root cause analysis, optimization suggestions, performance trend interpretation



Scheduling and Alerting System

- Cron Jobs $\Rightarrow$ Scheduling Test Runs
- Nodemailer $\Rightarrow$ Sends Alerts to Users
- Every hour, week, month, or 6 months



A screenshot of a 'SCHEDULE & ALERTS' configuration modal. The modal has a title bar with 'SCHEDULE & ALERTS' and a close button. It is divided into two main sections: 'TEST SCHEDULE' and 'REGRESSION ALERTS'. Under 'TEST SCHEDULE', there is a dropdown menu currently set to 'Every Hour'. Under 'REGRESSION ALERTS', there is a text input field for 'Email Address' containing 'paiman.nejrabi2000@gmail.com'. Below this, there is a label 'Alert when metrics regress by more than:' followed by a '+ Add Metric' button. A row below shows a dropdown set to 'Latency', a text input with '10', and a '%' symbol. At the bottom right, there are two buttons: 'CANCEL' and 'SAVE CONFIGURATION'.

SCHEDULE & ALERTS

TEST SCHEDULE

Every Hour

REGRESSION ALERTS

Email Address

paiman.nejrabi2000@gmail.com

Alert when metrics regress by more than: [+ Add Metric](#)

Latency 10 %

[CANCEL](#) [SAVE CONFIGURATION](#)

Example Alert Email



Performance Regression Alert

Detected Regressions:

- throughput: -99.7% (threshold: 1%)

Test Details:

Timestamp: 5/22/2026, 3:14:16 PM

Version: N/A

Duration: N/Ams

Success Rate: 10000.0%

Current Metrics:

Latency:	131.61ms
Consensus:	131.20ms
Throughput:	7.60 r/s
Success Rate:	10000.0%

This alert was triggered by your ResilientDB performance monitoring system.
View the dashboard for more details and historical trends.

Demo

Future Advancement

- Automated Pull Request Testing
- Enhanced AI analysis with multi-model comparison capabilities
- Multi-Configuration Scheduled Testing

Conclusion

Thank you!